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Style over Content: Understanding User Satisfaction with AI Companions

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ABSTRACT

AI companions are socially-oriented chatbots designed and trained to engage in casual dialogue, provide role-playing experiences, and offer companionship. However, there is a dearth of research on how these AI companions can be designed to meet user needs. The loneliness pandemic signifies a shift in user preferences from efficiency and utility towards hedonism and socialization. Given the diverse and flexible ways AI companions can communicate with users, the communication style of AI companions plays an instrumental role in meeting users' psychological needs. Consequently, building on the social style model, we construct a communication style framework tailored for AI companions and explore how its constituent dimensions of assertiveness, emotiveness, and imaginativeness satisfy users' social and hedonic needs. Preliminary analysis on a leading social media platform indicates that communication styles of AI companions can satisfy both users' hedonic and social needs. Expected contributions to theory and practice are then discussed.

Keywords

AI companion, communication style, social needs, hedonic needs, user satisfaction.

INTRODUCTION

The advent of Generative AI (GenAI) has empowered chatbots to engage in human-like conversations. Although the application of GenAI chatbots has been primarily task-driven to-date, socially-oriented chatbots are becoming prominent amid the current "loneliness pandemic" (Metz 2020). Individuals are increasingly turning to these chatbots to mitigate feelings of anxiety, boredom, or loneliness (Pentina et al. 2023). Chaturvedi et al. (2023) labelled these socially-oriented chatbots as AI companions,

GenAI characters designed and trained to engage in casual dialogue, offer companionship, and/or provide role-playing experiences. Unlike their task-oriented counterparts which are devised as conversational interfaces for service provision, AI companions possess unique personas, backstories, and conversational corpora with distinctive styles, aiming to foster engaging interactions and captivate users through immersive dialogue (Shum et al. 2018).

Despite the growing popularity of AI companions, the design principles necessary to meet user needs remain underdeveloped. Researchers have identified a myriad of psychological needs that users seek to fulfill through task-oriented chatbots, including utilitarian, technology, hedonic, and social gratification (Xie et al. 2024). However, these needs are not fully applicable to the context of AI companions. The loneliness pandemic signifies a shift in user preferences from functional utility towards social interaction and pleasure when engaging with GenAI chatbots for companionship. While prior research primarily focuses on the use of task-oriented chatbots, how AI companions meet users' hedonic and social needs remains unclear.

Given the diverse and flexible ways AI companions can communicate with users, a significant gap exists in the design principles for tailoring their communication styles to meet users' psychological needs. Research on the impact of communication styles on user satisfaction in the context of service chatbots has yielded mixed findings. For instance, positive emotions from AI can diminish satisfaction when they contradict user expectations, while sympathetic communication is sometimes preferred in service recovery (Han et al. 2023). Given that AI companions are primarily focused on the interaction experience and relationship development with users, this study posits that users would favor a more socially-oriented

communication style for emotional engagement and entertainment.

An extensive body of research on interpersonal social interaction has attested to the critical role of communication styles in fulfilling individuals' psychological needs across various contexts (Buller and Buller 1987). The social style model (Snaveley and McNeill, 2008) identifies two primary dimensions of communication style: assertiveness, the degree to which an individual attempts to influence others, and emotiveness, the expression of emotions during interactions. A unique attribute of AI companions, in contrast to human counterparts, stems from their adaptable communication styles. This adaptability enables the personalization of interactions to better meet users' needs. Furthermore, AI companions can engage in communication styles that are unconventional or even impossible in typical social interactions.

To capture these distinctive communicative traits, we introduce a novel dimension to the existing model: imaginativeness. This dimension encompasses the ability to stimulate imagination and transform fantasies into reality (Zagacki et al. 1992). In this sense, we delve into three key dimensions that, when taken together, constitute the communication style of AI companions: assertiveness, emotiveness, and imaginativeness. We endeavor to assess how each of these dimensions influences users' experiences and meets social needs or hedonic needs when conversing with AI companions. Overall, our research addresses the research question: How do the communication styles of AI companions facilitate the satisfaction of users' social and hedonic needs?

We construct a research model to formulate the impact of AI companion communication styles on user need gratification. To validate our hypotheses, we collected a large dataset from Reddit, comprising 10,449 posts by 8,956 users sharing screenshots of their dialogues with AI companions. These screenshots were processed via Optical Character Recognition (OCR) before conducting econometric analysis to validate our hypotheses. We present our preliminary findings and discuss potential implications for theory and practice.

THEORETICAL BACKGROUND AND HYPOTHESIS DEVELOPMENT

How AI Companions Satisfy Users

Numerous scholars have studied how individuals gratify social and hedonic needs via human interactions (Gan and Li 2018), such as the effects of gratifications on users' continuance intentions (Ji and Wayne Fu 2013) and their effects on the loyalty of online game users (Huang and Hsieh 2011). Although utilitarian needs and technological needs are also crucial in shaping user experience in the service context, they are less emphasized by users in the context of AI companions. Instead, the social interactions and immersive experiences shaped by personalized

personas and backstories, which are beyond the capabilities of task-oriented chatbots, highlight social and hedonic needs as the two key dimensions that AI companions can satisfy.

Social Needs

According to Self-Determination Theory (SDT), optimal motivation and well-being are achieved when three basic social needs are satisfied: autonomy, competence, and relatedness (Vansteenkiste et al. 2010). This theory has been widely applied to understand users' motivations and psychological needs in chatbot interactions. For instance, Jiménez-Barreto et al. (2021) explored motivational factors driving chatbot usage and Xia et al. (2023) investigated the impact of anthropomorphic features in chatbots on user experiences. Within our research, the need for autonomy is defined as users' desire for control over conversations with AI companions; the need for competence as the ability to derive self-worth and boost self-esteem through interactions; and the need for relatedness as the desire to establish an emotional connection with AI companions (Nikou and Economides 2017). These needs collectively form the social framework users seek to satisfy through their interactions with AI companions.

Satisfaction is measured by the extent to which users are pleased with their interactions with AI companions (Nguyen et al. 2022). A user's satisfaction with an AI companion hinges on the fulfillment of both their social and hedonic needs. Specifically, when users' need for autonomy is met, their actions become self-determined and aligned with their interests, enhancing their overall satisfaction. Similarly, when the need for competence is satisfied, it leads to increased self-worth and satisfaction. Moreover, fulfilling the need for relatedness heightens satisfaction through emotional connections and feelings of care. Therefore, we propose:

Hypothesis 1a, b, c: The gratification of users' social needs for (a) autonomy, (b) competence, and (c) relatedness is positively associated with satisfaction with an AI companion.

Hedonic Needs

Hedonic needs involve the desire for fun, pleasure, and escapism when interacting with AI companions (Odağ et al. 2016). AI companions, through their distinctive personas, engaging backstories, and versatile communication styles, provide entertaining experiences that beyond the social and utilitarian. Users often look to AI companions for help in passing time and temporarily escaping real-life issues, adding an element of escapism to their interactions (Cheng and Jiang 2020). The enjoyment derived from interacting with an AI companion can substantially enhance overall user satisfaction. Thus, we propose:

Hypothesis 2: The gratification of users' hedonic needs is positively associated with satisfaction with an AI companion.

Communication Styles of AI Companions

Communication style refers to consistent conversational patterns during interactions (Snavely and Walters, 1983). The social style model (Snavely and McNeill 2008), identifies two main dimensions: assertiveness, which indicates how certain and clear a person is, and emotiveness, which reflects their emotional expressiveness. Both dimensions can determine whether individuals satisfy their psychological needs through conversation. Given that communication styles are relevant both online and offline (Derks et al. 2008), and considering the similarities between interactions with AI companions and humans, interpersonal communication styles can be effectively applied to our research context.

While assertiveness and emotiveness are common dimensions used to assess communication styles, they do not fully capture the unique traits of AI companions. Unlike humans, AI can engage in imaginative and unconventional dialogues, free from the constraints of identities and social norms (McDonald and Crandall 2015). Thus, this study introduces imaginativeness as a novel dimension, referring to the extent to which an AI companion engages in creative dialogue. This allows AI companions to provide unique social experiences that go beyond traditional communication, immersing users in a fictional world through dialogue.

The assertive communication style plays a crucial role in fulfilling users' social needs through interactions with AI companions. Firstly, it helps users feel in control by providing clear and unambiguous responses, allowing them to direct the conversation as they desire. Furthermore, it enables users to perceive AI companions as entities with coherent thoughts and perspectives, rather than as preprogrammed chatbots (Yen & Chiang, 2021). This sense of control can satisfy users' need for autonomy. Additionally, decisive and clear responses from AI companions can affirm users' thoughts, impart a sense of respect during the interaction, and boost confidence and self-esteem (Omura et al., 2017), fulfilling the need for competence. Lastly, AI companions demonstrate active listening and empathy through clear communication of their thought processes and direct expression, which strengthens users' connection with the AI companion and facilitates the gratification of the need for relatedness (Winer et al., 2024). Therefore, we propose:

Hypothesis 3a, b, c: The assertiveness of an AI companion can facilitate the gratification of users' social needs for (a) autonomy, (b) competence, and (c) relatedness.

The emotive communication style is crucial for fulfilling users' social needs. According to the emotions as social information model (Van Kleef, 2009), the emotions expressed by AI companions provide social information

that conveys their emotional reactions evoked by users, which often make users aware of their ability to influence others, thus satisfying their need for autonomy. Additionally, emotiveness enhances the impact of communication, making users feel valued by AI companions. Such instances where AI companions express strong emotions are considered rare, and when users successfully get a strong emotional response from AI, it can boost their sense of competence (Gelbrich et al., 2021). Furthermore, the emotional expressiveness of chatbots can make them appear more authentic, thereby fostering an emotional connection with users (Han et al., 2022). Positive emotions expressed by an AI companion can provide emotional support (Ta et al., 2020), while negative emotions can be perceived as signs of authenticity and attachment to the relationship, thereby strengthening users' emotional investment (Sanford & Rowatt, 2004). Based on these considerations, we propose:

Hypothesis 4a, b, c: The emotiveness of an AI companion can facilitate the gratification of users' social needs for (a) autonomy, (b) competence, and (c) relatedness.

We propose imaginativeness as a crucial enabler of users' hedonic experiences with AI companions, drawing on the insight that daydreaming often provides greater enjoyment than the realization of those dreams (Heath & Nixon, 2021). Imaginativeness offers an eccentric and unconventional mode of conversation, capable of eliciting more imaginative and unexpected responses compared to traditional communication styles, which stimulates excitement and enhances enjoyment (McCann & Honeycutt, 2006). Furthermore, imaginativeness fosters a captivating and immersive experience, enabling users to escape from the real world and satisfy their hedonic needs. Therefore, we propose:

Hypothesis 5: The imaginativeness of an AI companion can facilitate the gratification of users' hedonic needs.

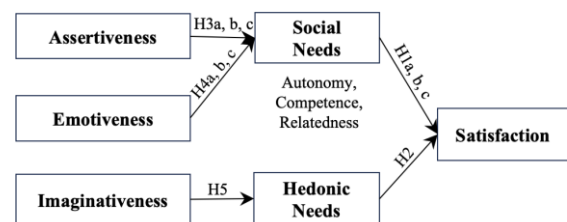


Figure 1. Research Model

METHODOLOGY

Data Collection

Character.AI is one of the most popular AI companion applications, which was named Google Play's AI App of 2023. It allows users to create and talk to advanced AI companions, including language tutors, text adventure

games, life advisors, and more. Our research used Character.AI as an empirical context of AI companions.

Our dataset was collected from Reddit, a popular topic-driven social media platform. We collected posts made by members in the “Character.AI” subreddit, which is established in August 2022, for sharing screenshots of their conversation with AI companions. Each post consists of a title and a short description. Our dataset contains all 10,449 posts with screen captures made by 8,956 users in 2023.

Variable Operationalization

To conduct a preliminary analysis, we firstly employed the Optical Character Recognition (OCR) toolkit, EasyOCR, to extract the textual content of users’ conversations with AI companions from screen captures. It is an open-source OCR engine leveraging the Character Region Awareness for Text detection (CRAFT) algorithm for detection (Baek et al., 2019) and the Convolutional Recurrent Neural Network (CRNN) for text recognition (Shi et al., 2015).

We identified each user’s satisfaction level expressed in the title and description of each post using Zero-Shot Learning (ZSL) (Pourpanah et al., 2023; Xian et al., 2018). Specifically, we employed the DeBERTa-v3-base-mnli-fever-anli model, which was trained on the MultiNLI, FEVER-NLI, and Adversarial-NLI datasets, comprising 763,913 hypothesis-premise pairs for natural language inference (Laurer et al., 2024).

We then identified three communication styles of AI companions based on the messages they post in their conversations with each user. We used ZSL to identify the level of assertiveness. We operationalized emotiveness as the overall sentiment intensity computed through the Valence Aware Dictionary for eSentiment Reasoning (VADER), a lexicon and rule-based sentiment analysis tool specifically attuned to nuances in social media text (Hutto & Gilbert, 2014). We calculated imaginativeness by using a fine-tuned version of the MiniML-L12-H384-uncased model from Wang et al. (2020) to estimate the degree of fictionality for each conversation.

We captured the social needs and hedonic needs expressed by each user in their screen-captured conversation with AI companions. First, we utilized ZSL to calculate scores for the sense of control, self-esteem, and social status. Then, using a fine-tuned roberta-base model trained on the go_emotions dataset, which contains 58k carefully curated Reddit comments labeled for 28 emotion categories (Demszky et al., 2020), we calculated intensity scores for emotions including pride, love, admiration, gratitude, caring, joy, amusement, relief, and curiosity. Afterward, we estimated a user’s need for autonomy with the score for the sense of control. We combined the scores for pride, self-esteem, and social status to estimate a user’s need for competence. Likewise, we combined the scores for love, admiration, gratitude, and caring to represent a user’s need for relatedness. Lastly, we combined the scores for joy,

amusement, relief, and curiosity to indicate a user’s need for hedonism.

Data Analysis

Table 1 summarizes the model estimation results produced by our Ordinary Least Squares regression. Model 3 shows that both social needs for autonomy ($\beta = 0.3229$, $p < 0.001$), competence ($\beta = 0.7313$, $p < 0.001$), relatedness ($\beta = 1.5979$, $p < 0.001$), and hedonic needs ($\beta = 0.1214$, $p < 0.01$) were positively associated with user satisfaction, supporting both Hypothesis 1 and 2. Model 4 and 5 showed that assertiveness was positively associated with the need for autonomy ($\beta = 0.0304$, $p < 0.01$) and competence ($\beta = 0.0165$, $p < 0.01$), supporting Hypothesis 3a and 3b. Model 4, 5 and 6 showed that emotiveness was positively associated with the need for autonomy ($\beta = 0.0108$, $p < 0.05$), competence ($\beta = 0.0092$, $p < 0.01$), and relatedness ($\beta = 0.0026$, $p < 0.01$), supporting Hypothesis 4. Model 7 showed that imaginativeness was positively associated with hedonic needs, supporting Hypothesis 5. However, Model 6 showed that the impact of assertiveness on the need for relatedness was not significant, and Hypothesis 3c was not supported. One possible explanation for this surprising finding is that the assertive communication style can appear cold and apathetic, whereas emotional expressiveness is key to gratifying the need for relatedness.

Additionally, as shown in Table 2, our results strongly support the mediating effects of psychological needs. Specifically, we found that the need for autonomy mediates the impact of assertiveness on satisfaction ($\beta = 0.018***$); the need for competence mediates the impact of assertiveness ($\beta = 0.013***$) and emotiveness ($\beta = 0.008**$) on satisfaction; and the need for relatedness mediates the impact of emotiveness on satisfaction ($\beta = 0.004**$). However, the results showed that the indirect impact of emotiveness through the need for autonomy on satisfaction is not significant ($\beta = 0.005n.s.$). This finding shows that the satisfaction users derived from conversing with an emotive AI companion can be attributed to a heightened sense of competence and relatedness, rather than an enhanced sense of control over the AI companion’s responses. Since Hypothesis 3c is not supported, and imaginativeness does not significantly impact satisfaction in Model 2, the mediation test results for the indirect effect of assertiveness on satisfaction through the need for relatedness, and that of imaginativeness on satisfaction through hedonic needs, are not presented.

EXPECTED CONTRIBUTIONS AND DIRECTIONS OF FUTURE RESEARCH

The expected contributions of this study to the field are significant in several areas. First, we integrate insights from the interpersonal literature with the unique characteristics of AI companions to establish a framework of communication styles geared towards gratifying users’ psychological needs. While previous research has delineated various attributes of chatbot communication

(Yang et al., 2023), the influence of communication styles on users' interactive experiences with AI companions remains underexplored. To address this gap, our study expands the model by incorporating a novel dimension, imaginativeness, alongside assertiveness and emotiveness. This improvement advances AI companion research by highlighting both their similarities and differences from human interaction.

Furthermore, our findings offer practical guidelines for designing AI companions. Emphasizing the importance of communication style, our research suggests that AI companion platforms should allow users to specify preferred communication styles during the creation and configuration of an AI companion, in addition to providing a training corpus. Our results indicate that the communication style is as impactful to users' experience, if not more, as the content of the dialogue itself. Moreover,

AI companion platforms can leverage these insights to personalize the communication style of each AI companion for different users to cater to their diverse needs.

This study can be enriched in the following directions. First, the operationalization of variables in this study is relatively preliminary, so more sophisticated methods should be employed to enhance the validity of our measurement instruments. Second, since users may have preferred communication styles when interacting with AI companions, future research could explore the alignment of communication styles from both sides. Third, this study relied on objective data to examine the gratification of users' psychological needs through conversations with AI companions. Additional experiments can be conducted to provide deeper insights into users' subjective experiences with AI companions.

Variable	Satisfaction with an AI Companion			Social Needs			Hedonic Needs
	M1	M2	M3	Autonomy	Competence	Relatedness	M7
Constant	0.3121 ^{n.s.} (0.119)	0.2677 ^{n.s.} (0.181)	-0.2059 ^{n.s.} (0.149)	0.6290** (0.001)	0.2032* (0.042)	0.0814* (0.013)	0.0092 ^{n.s.} (0.818)
Independent Variables							
Assertiveness		0.0280** (0.006)	0.0088 ^{n.s.} (0.227)	0.0304** (0.002)	0.0165** (0.001)	-0.0020 ^{n.s.} (0.233)	-0.0003 ^{n.s.} (0.901)
Emotiveness		0.0265*** (0.000)	0.0123** (0.001)	0.0108* (0.048)	0.0092** (0.001)	0.0026** (0.004)	0.0005 ^{n.s.} (0.675)
Imaginativeness		0.0019 ^{n.s.} (0.815)	-0.0004 ^{n.s.} (0.948)	-0.0020 ^{n.s.} (0.800)	0.0042 ^{n.s.} (0.306)	-0.0004 ^{n.s.} (0.750)	0.0035* (0.034)
Mediating Variables							
Social Needs	Autonomy		0.3229*** (0.000)				
	Competence		0.7313*** (0.000)				
	Relatedness		1.5979*** (0.000)				
Hedonic Needs			0.1214** (0.001)				
Observations	10449	10449	10449	10449	10449	10449	10449
Adjusted R^2	0.046	0.048	0.516	0.019	0.030	0.015	0.040
F	11.83	11.86	254.4	5.133	7.532	4.144	9.879

Control Variables: Duration_posts, number_subreddites, word_count, lexical_density, lexical_diversity, sentiment_subjectivity, age, theme, created_time, type_screenshot, flair

Table 1. OLS Results

Relationship	β	95% CI	P-value	RME (%)
Assertiveness → Need for autonomy → Satisfaction	0.016	(0.006, 0.027)	0.006	55.89
Assertiveness → Need for competence → Satisfaction	0.010	(0.004, 0.017)	0.002	54.47
Emotiveness → Need for autonomy → Satisfaction	0.006	(-0.000, 0.014)	0.056	24.21
Emotiveness → Need for competence → Satisfaction	0.007	(0.003, 0.012)	0.006	32.11
Emotiveness → Need for relatedness → Satisfaction	0.004	(0.001, 0.007))	0.008	27.85

Notes: β = path coefficients, 95% CI = 95% confidence interval, RME = Relative mediation effect

Table 2. Mediating Effect Results

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